

Basic Surveying - Important Questions

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- ① Define surveying?
- ② State two main principles of surveying.
- ③ What is a plane survey? Mention where is it used.
- ④ What is a geodetic survey? Mention where is it used.
- ⑤ Define cadastral survey. State what it is used for.
- ⑥ What is meant by working "whole to part" in surveying?
- ⑦ Why is curvature of the earth ignored in plane surveying?
- ⑧ Which survey is suitable for small areas? Justify your answer.
- ⑨ Differentiate between plane surveying and geodetic surveying. (at least 2 points)
- ⑩ Why are topographical surveys important?
- ⑪ What is the object of surveying?
- ⑫ Name two instruments used in chain survey.
- ⑬ What is the difference between open traverse & closed traverse?
- ⑭ Explain briefly about plane table surveying.
- ⑮ Define the following:
a) Map Scale,
b) Representative Fraction (R.F.).
- ⑯ Convert a scale of 1 cm = 500 m into R.F.
- ⑰ A map has an R.F. of $\frac{1}{2500}$. If the R.F. is converted to a verbal scale, what would the scale be?
- ⑱ If 5 cm on a map represents 400 m on the ground, find the actual area of a plot on ground that measures 6 cm² on the map. Give your answer in square meters.
- ⑲ Why should units be same in R.F. (Representative Fraction) calculations?
- ⑳ State the uses of surveying. (any four points)
- ㉑ State the uses of surveying.

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UNIT 1: INTRODUCTION TO SURVEYING

Define surveying. What are its objectives?

Types of surveying – explain plane and geodetic surveying.

Principles of surveying.

Classification of surveying based on instruments and methods.

What is scale? Types of scales used in surveying.

Define accuracy and precision in surveying.

Difference between mapping and surveying.

Uses of surveying in civil engineering.

What are errors? Types of errors in surveying.

Chain surveying – basic concept.

- ◆ Short Questions

What is a survey station?

Define baseline.

What is ranging?

What is offset?



UNIT 2: CHAIN SURVEYING

- ♦ Important Questions

Explain chain surveying with neat sketch.

Instruments used in chain surveying.

Types of chains and their lengths.

Ranging – direct and indirect ranging.

Offsets – types and methods.

Obstacles in chaining and how to overcome them.

Errors in chain surveying.

Booking of field notes.

Advantages and limitations of chain surveying.

Chain survey vs compass survey.

- ♦ Short Questions

What is a cross staff?

Define tie line.

What is check line?



UNIT 3: COMPASS SURVEYING

- ◆ Important Questions

| Principle of compass surveying.

Types of compasses – prismatic and surveyor's compass.

Bearings – whole circle bearing and reduced bearing.

Conversion of bearings (WCB $\leftrightarrow$ RB).

Local attraction – causes and correction.

Magnetic declination – definition and types.

Traverse surveying using compass.

Errors in compass surveying.

Difference between true bearing and magnetic bearing.

Plotting of compass survey.



UNIT 4: LEVELLING

♦ Important Questions

Define levelling and its objectives.

Types of levelling.

Dumpy level – parts and functions.

Levelling staff – types.

Terms used in levelling (RL, BM, HI, etc.).

Methods of levelling:

Height of Instrument method

Rise and Fall method

Arithmetic checks in levelling.

Errors in levelling and precautions.

Profile levelling and cross-sectioning.

Temporary and permanent adjustments of level.